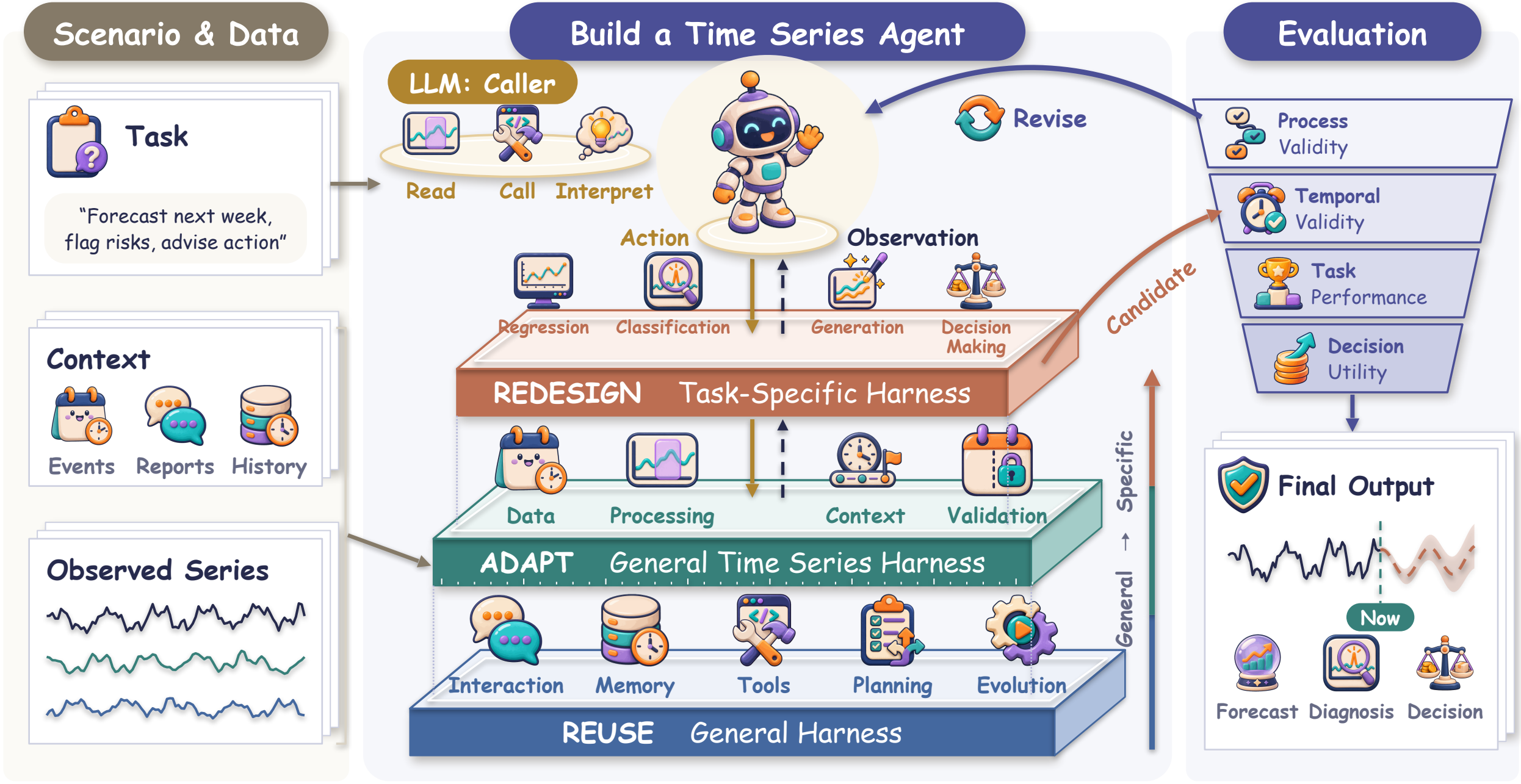
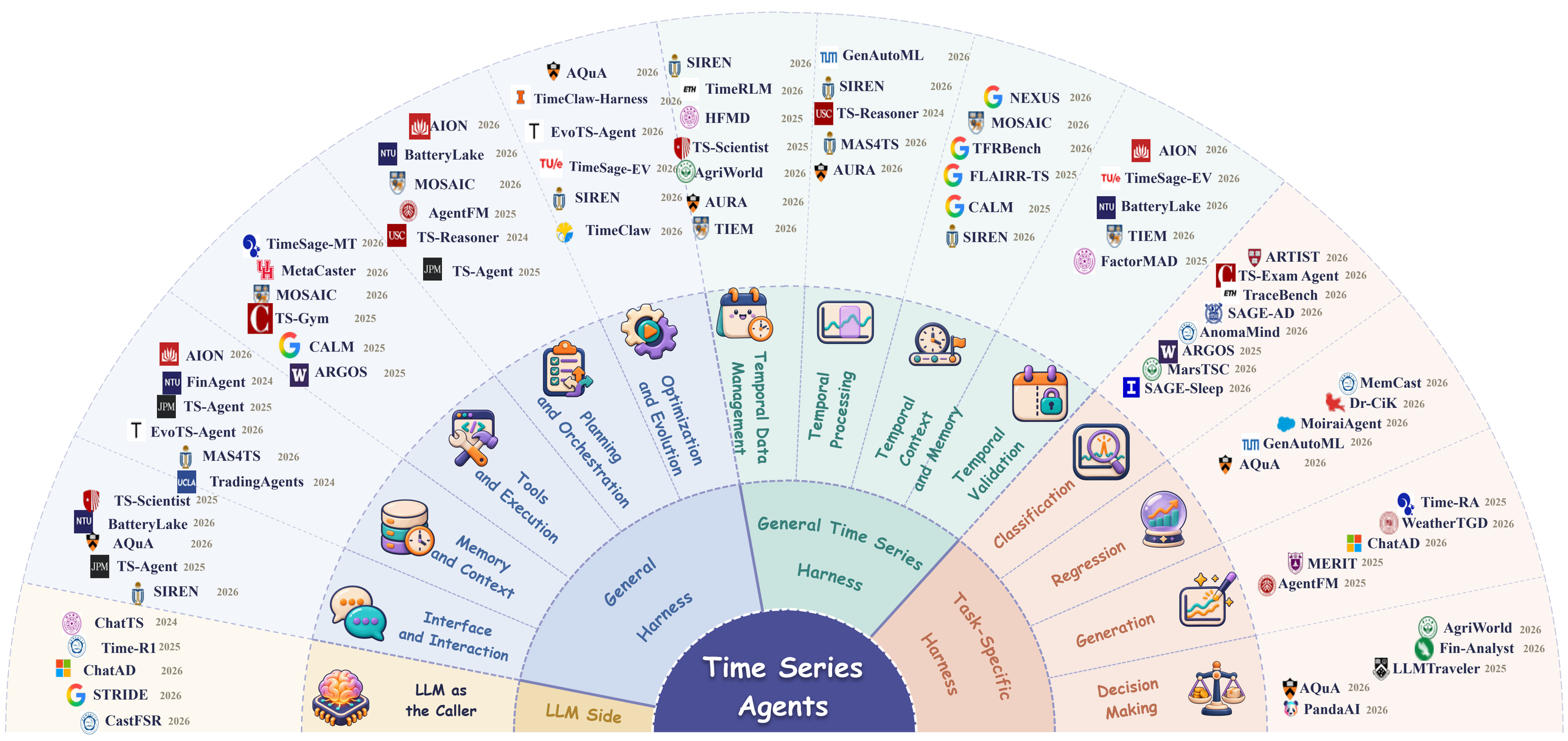
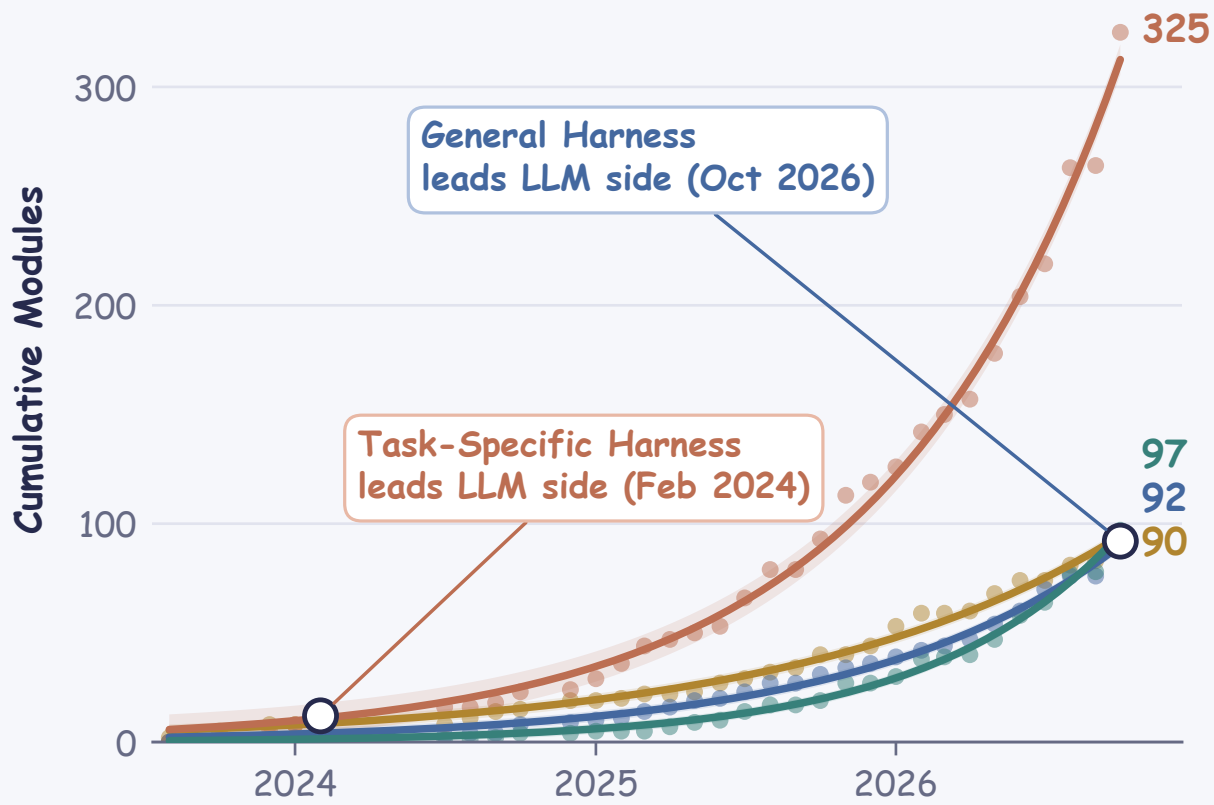

Scenario & Data
Task
"Forecast next week, flag risks, advise action"
Context
Events
Reports
History
Observed Series
Build a Time Series Agent
LLM: Caller
Read
Call
Interpret
Revise
Action
Observation
Regression
Classification
Generation
Decision Making
REDESIGN Task-Specific Harness
Data
Processing
Context
Validation
ADAPT General Time Series Harness
Interaction
Memory
Tools
Planning
Evolution
REUSE General Harness
Specific
General
Candidate
Evaluation
Process Validity
Temporal Validity
Task Performance
Decision Utility
Final Output
Now
Forecast
Diagnosis
Decision

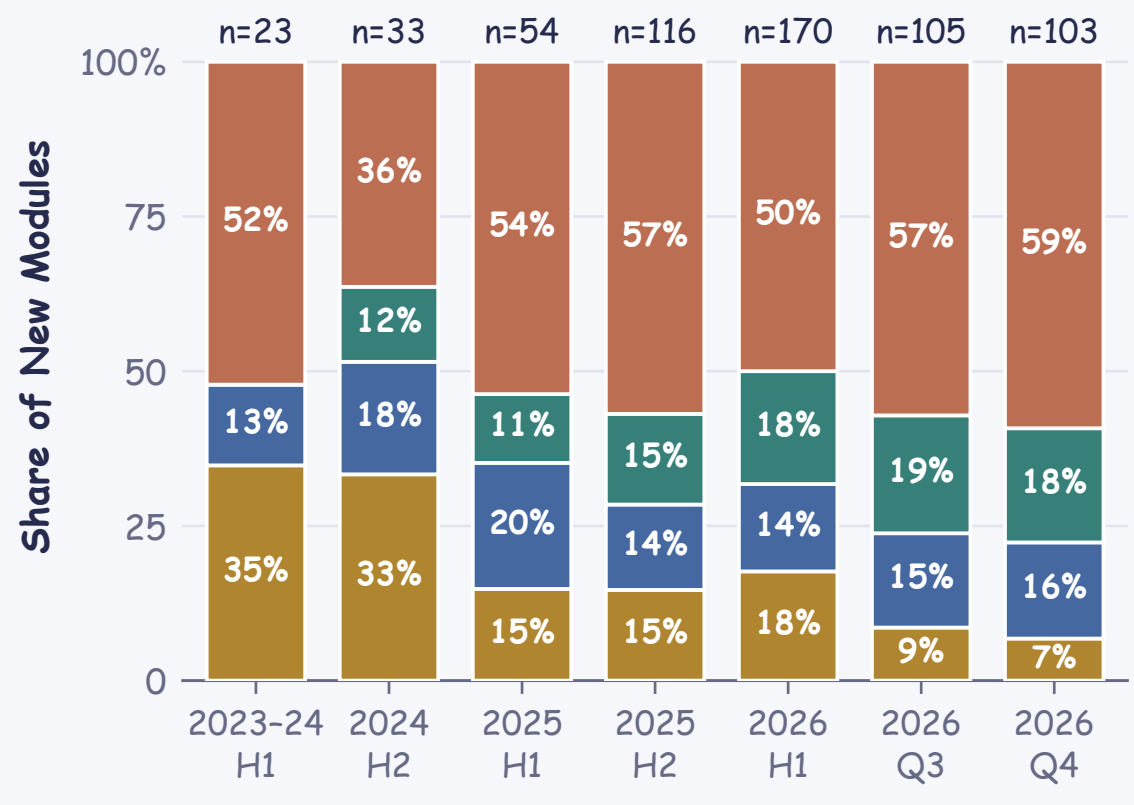

Time Series Agents
LLM Side
LLM as the Caller
General Harness
Interface and Interaction
Memory and Context
Tools and Execution
Planning and Orchestration
Optimization and Evolution
General Time Series Harness
Temporal Data Management
Temporal Processing
Temporal Context and Memory
Temporal Validation
Task-Specific Harness
Classification
Regression
Generation
Decision Making
ChatTS 2024
Time-R1 2025
ChatAD 2026
STRIDE 2026
CastFSR 2026
TS-Scientist 2025
BatteryLake 2026
AQuA 2026
TS-Agent 2025
SIREN 2026
AION 2026
FinAgent 2024
TS-Agent 2025
EvoTS-Agent 2026
MAS4TS 2026
TradingAgents 2024
TimeSage-MT 2026
MetaCaster 2026
MOSAIC 2026
TS-Gym 2025
CALM 2025
ARGOS 2025
AQuA 2026
TimeClaw-Harness 2026
AION 2026
BatteryLake 2026
MOSAIC 2026
AgentFM 2025
TS-Reasoner 2024
EvoTS-Agent 2026
TimeSage-EV 2026
SIREN 2026
TimeClaw 2026
TS-Agent 2025
SIREN 2026
TimeRLM 2026
HFMD 2025
TS-Scientist 2025
AgriWorld 2026
AURA 2026
TIEM 2026
GenAutoML 2026
SIREN 2026
TS-Reasoner 2024
MAS4TS 2026
AURA 2026
NEXUS 2026
MOSAIC 2026
TFRBench 2026
FLAIRR-TS 2025
CALM 2025
SIREN 2026
AION 2026
TimeSage-EV 2026
BatteryLake 2026
TIEM 2026
FactorMAD 2025
ARTIST 2026
TS-Exam Agent 2026
TraceBench 2026
SAGE-AD 2026
AnomaMind 2026
ARGOS 2025
MarsTSC 2026
SAGE-Sleep 2026
MemCast 2026
Dr-CiK 2026
MoiraiAgent 2026
GenAutoML 2026
AQuA 2026
Time-RA 2025
WeatherTGD 2026
ChatAD 2026
MERIT 2025
AgentFM 2025
AgriWorld 2026
Fin-Analyst 2026
LLMTraveler 2025
AQuA 2026
PandaAI 2026

LLM Side (doubling 9.2 mo)
General Harness (doubling 7.1 mo)
General Time Series Harness (doubling 5.3 mo)
Task-Specific Harness (doubling 6.6 mo)

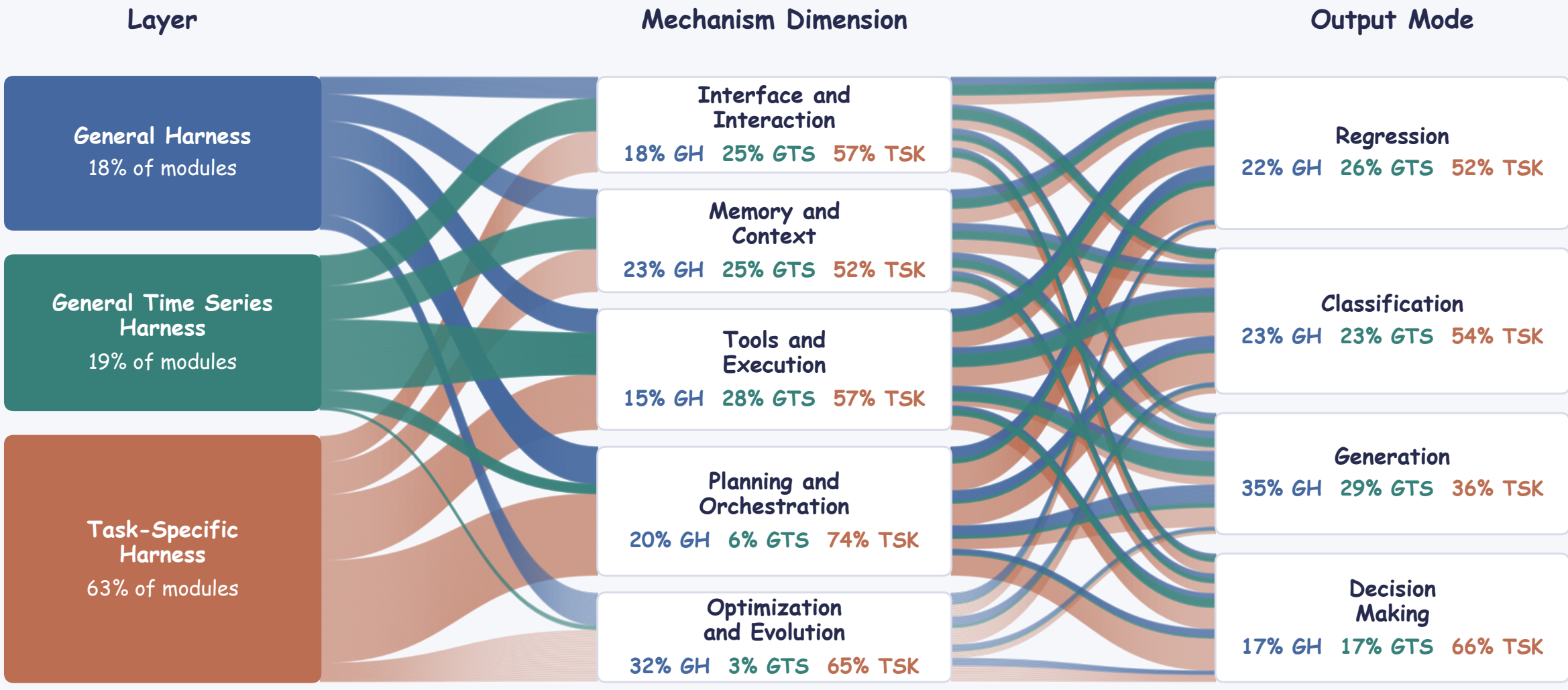
(a) Cumulative Module Contributions



(b) Share of New Modules by Period



(c) Harness Modules by Layer, Mechanism Dimension and Output Mode



(a) Output Modes × Temporal Functions

Column: Assemble

Row: Compare

	Regression	Classification	Generation	Decision Making
Temporal Data Management	History	Labeled Windows	Samples	State
Temporal Processing	Horizon	Intervals	Transforms and Views	Actions
Temporal Context and Memory	Events	Exemplars	References	Experience
Temporal Validation	Cutoff	Evidence	Fidelity and Claims	Constraints

(b) Assembling a Demand-Planning Agent by Cell

Execute Any Cell

General Harness

Prepare Any Cell

General Time Series Harness

Forecast Regression · After

History Now Forecast

Task-Specific Harness

Decide Decision · After

Task-Specific Harness

What Training Adds to the Caller

Reading Series in the Model



Series Patches → Language Space

Instructions and Outcome Feedback

Toward Native Perception



Series → Omni-Modal LLM

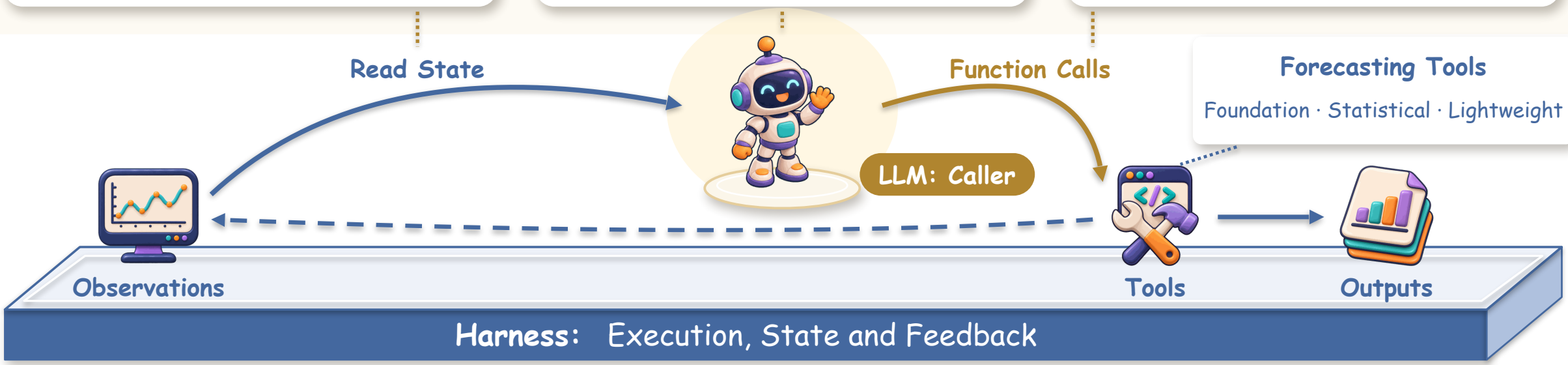
Outlook: Series as a Native Modality

Training the Caller



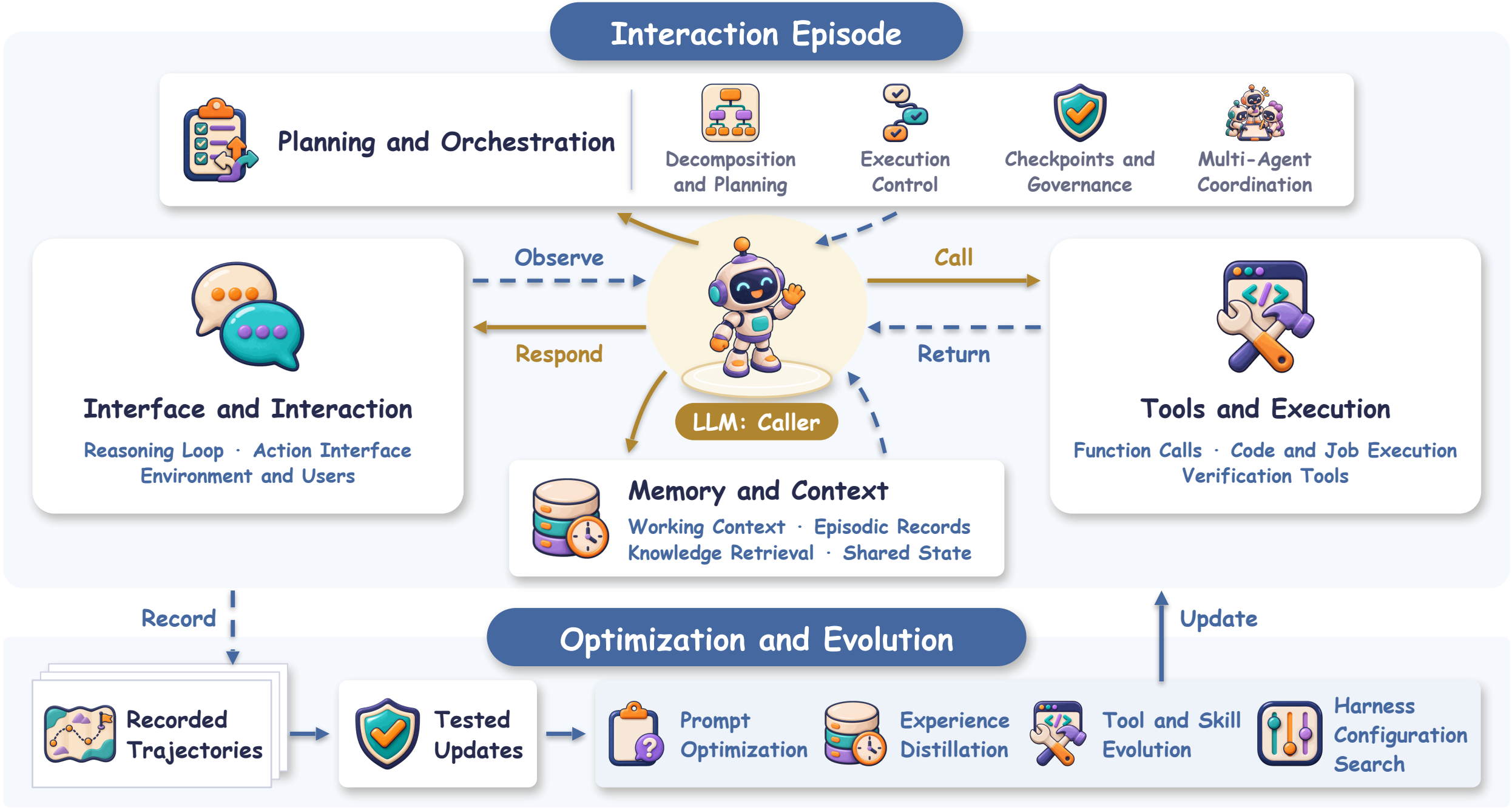
Tool Trajectories → Call Policy

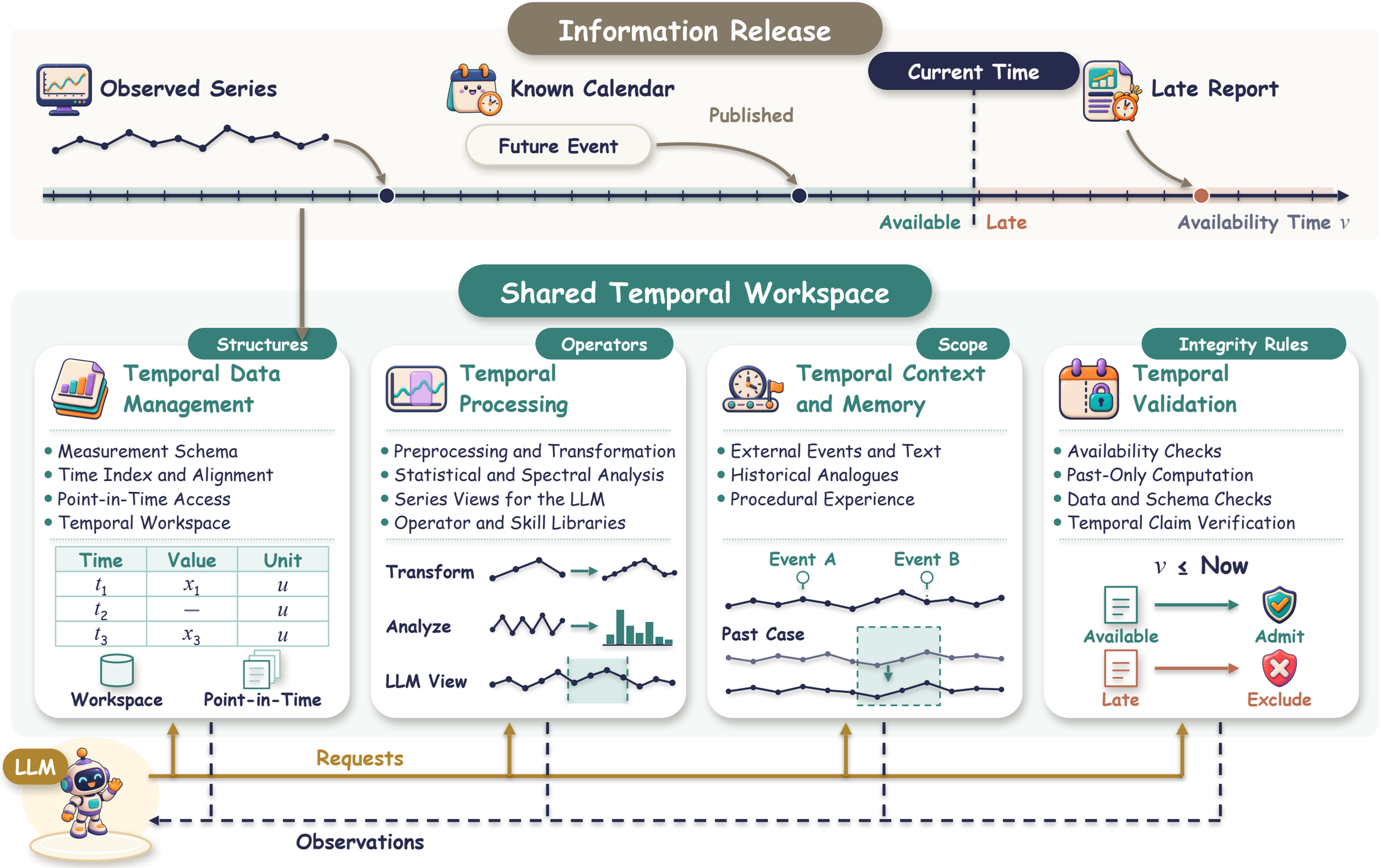
Execution and Preference Feedback



Evaluation Targets

- Forecast Accuracy
- Explanation Faithfulness
- Evidence Grounding
- Action Quality


Interaction Episode
Planning and Orchestration
Decomposition and Planning
Execution Control
Checkpoints and Governance
Multi-Agent Coordination
Observe
Call
Respond
Return
LLM: Caller
Interface and Interaction
Reasoning Loop · Action Interface
Environment and Users
Tools and Execution
Function Calls · Code and Job Execution
Verification Tools
Memory and Context
Working Context · Episodic Records
Knowledge Retrieval · Shared State
Record
Update
Optimization and Evolution
Recorded Trajectories
Tested Updates
Prompt Optimization
Experience Distillation
Tool and Skill Evolution
Harness Configuration Search

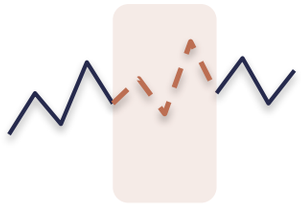

Information Release
Observed Series
Known Calendar
Current Time
Late Report
Published
Future Event
Available
Late
Availability Time v
Shared Temporal Workspace
Structures
Temporal Data Management
Measurement Schema
Time Index and Alignment
Point-in-Time Access
Temporal Workspace
Time
Value
Unit
t_1
x_1
u
t_2
—
u
t_3
x_3
u
Workspace
Point-in-Time
Operators
Temporal Processing
Preprocessing and Transformation
Statistical and Spectral Analysis
Series Views for the LLM
Operator and Skill Libraries
Transform
Analyze
LLM View
Scope
Temporal Context and Memory
External Events and Text
Historical Analogues
Procedural Experience
Event A
Event B
Past Case
Integrity Rules
Temporal Validation
Availability Checks
Past-Only Computation
Data and Schema Checks
Temporal Claim Verification
$v \leq$ Now
Available
Admit
Late
Exclude
LLM
Requests
Observations

Observed Record

Now

After the Current Time

Regression



Imputation and Estimation (15)

Imputation · Reconstruction · Estimation

Error Against Observed Values

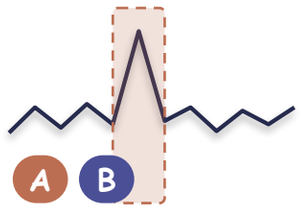


Value Forecasting (74)

Point and Probabilistic Forecasts

Error · Proper Scoring Rules

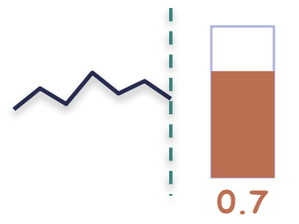
Classification



Classification and Detection (65)

Classification · Anomaly Detection
Root Cause · Closed Questions

Accuracy · F1 · AUC

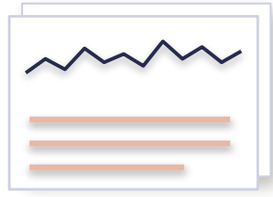


Event Forecasting (14)

Event Label · Event Probability

Accuracy · Brier Score

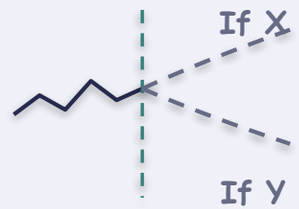
Generation



Description and Synthesis (40)

Open Questions · Explanation
Annotation · Synthesis

Quality · Fidelity · Use



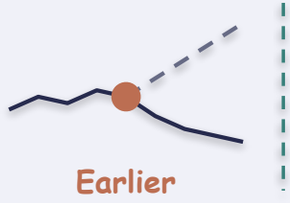
Projection

Open Problem

Futures Under Assumed Conditions

No Realized Future to Score

Decision Making

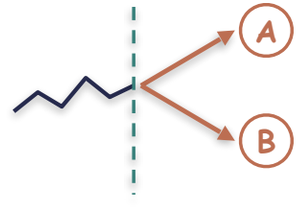


Hindsight Decision

Open Problem

Action That Should Have Been Taken

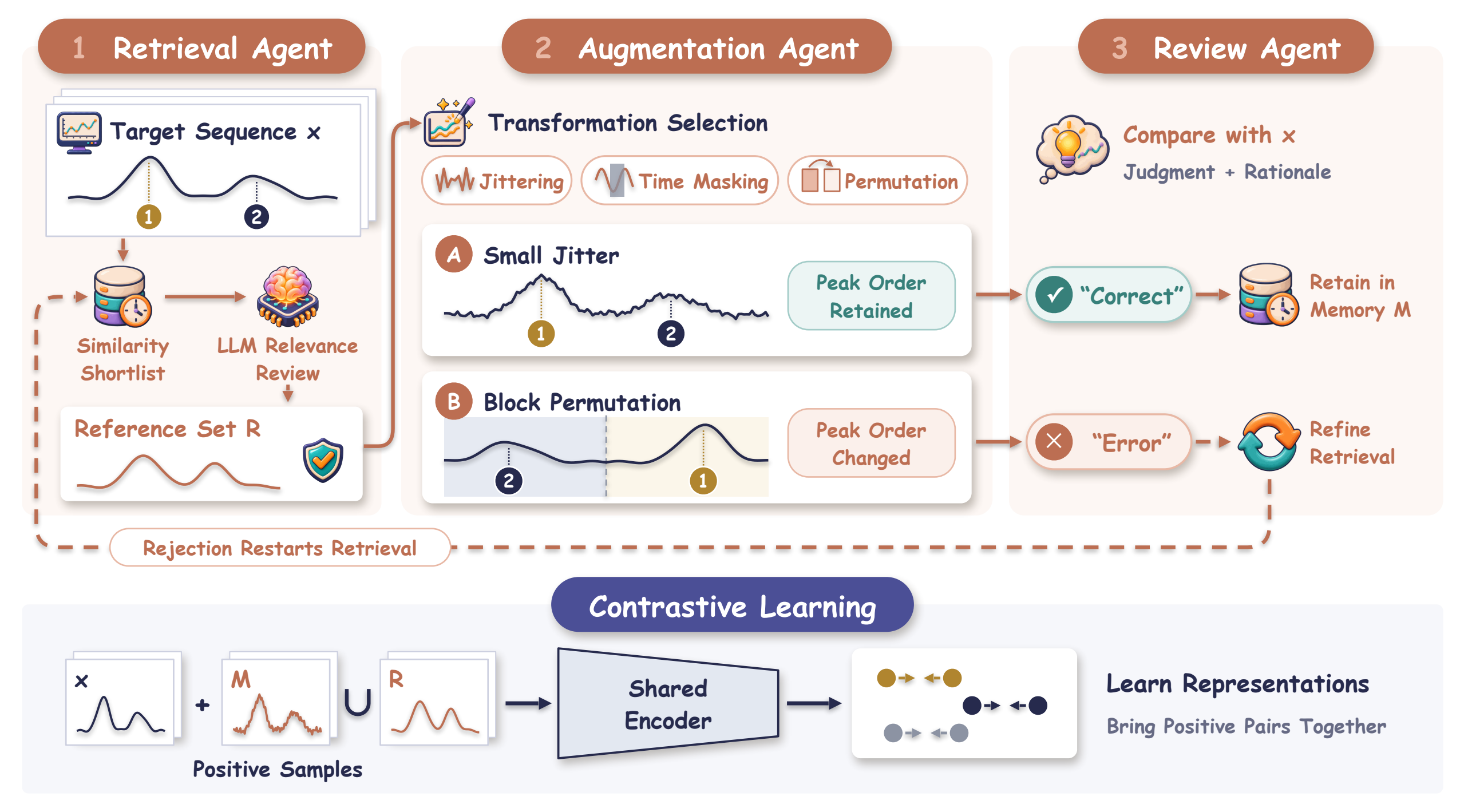
Replay · Counterfactual Estimate

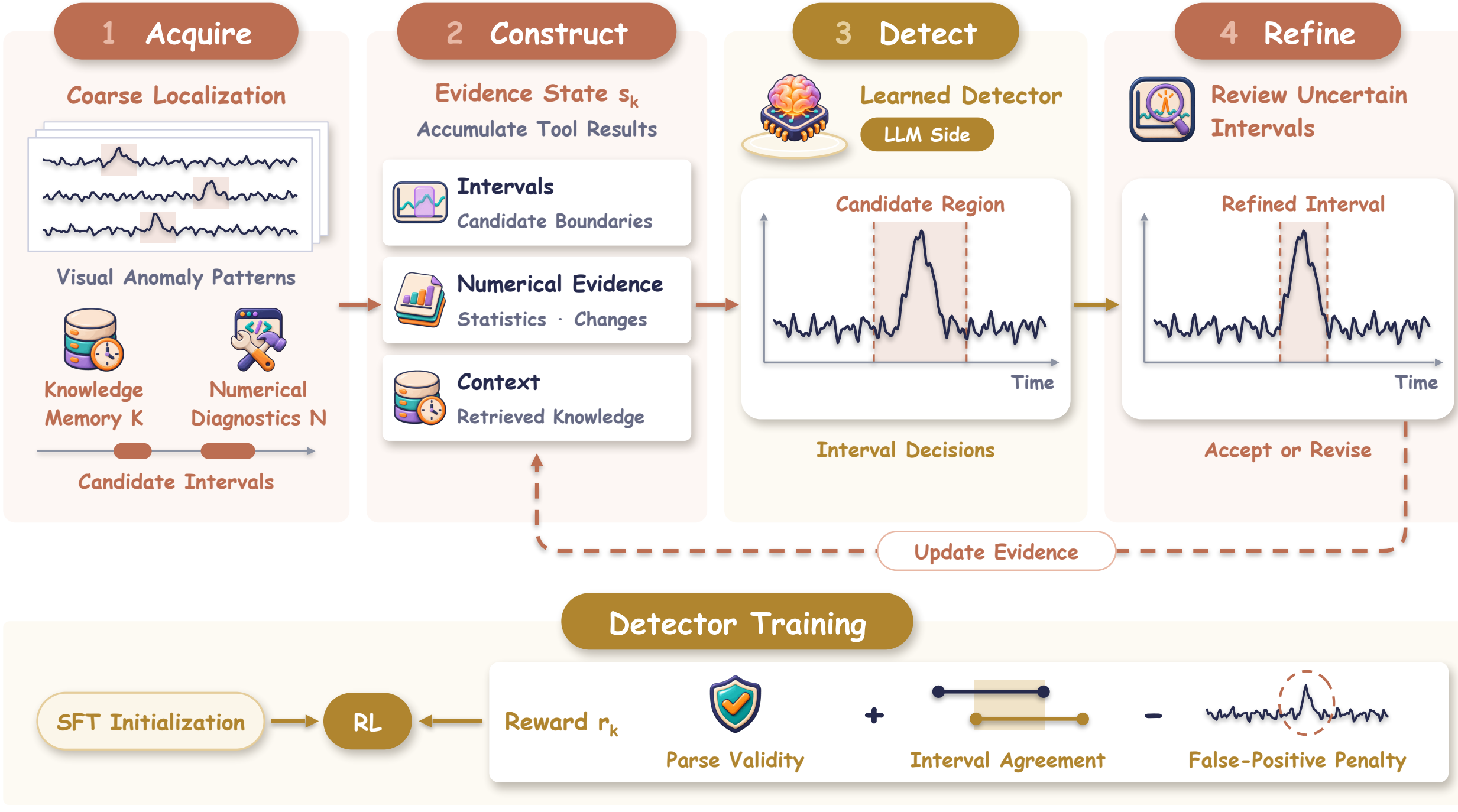


Decision (46)

Advisory Decision · Autonomous Control

Utility · Cost · Risk


1 Retrieval Agent
Target Sequence x
1
2
Similarity Shortlist
LLM Relevance Review
Reference Set R
2 Augmentation Agent
Transformation Selection
Jittering
Time Masking
Permutation
A Small Jitter
1
2
Peak Order Retained
B Block Permutation
2
1
Peak Order Changed
3 Review Agent
Compare with x
Judgment + Rationale
"Correct"
Retain in Memory M
"Error"
Refine Retrieval
Rejection Restarts Retrieval
Contrastive Learning
x
+
M
U
R
Positive Samples
Shared Encoder
Learn Representations
Bring Positive Pairs Together


1 Acquire
Coarse Localization
Visual Anomaly Patterns
Knowledge Memory K
Numerical Diagnostics N
Candidate Intervals
2 Construct
Evidence State s_k
Accumulate Tool Results
Intervals
Candidate Boundaries
Numerical Evidence
Statistics · Changes
Context
Retrieved Knowledge
3 Detect
Learned Detector
LLM Side
Candidate Region
Time
Interval Decisions
4 Refine
Review Uncertain Intervals
Refined Interval
Time
Accept or Revise
Update Evidence
Detector Training
SFT Initialization
RL
Reward r_k
Parse Validity
Interval Agreement
False-Positive Penalty

Evaluation Episode

Contract

Output Spec · Time Cutoff · Budget

Resources

Data · Code · Documents

Execution

Tool Call → Result / Failure

Artifact

Output File · Prediction (Now)

Evidence Links

Execution Record

Call	tool(input)
Status	Result or Error
Artifact	Output Reference

Resource Use

Latency · Cost · Tool Use

Artifact Validity

File — Complete, Readable

Schema — Required Structure

Execution — Runs, Loads or Parses

Task Performance



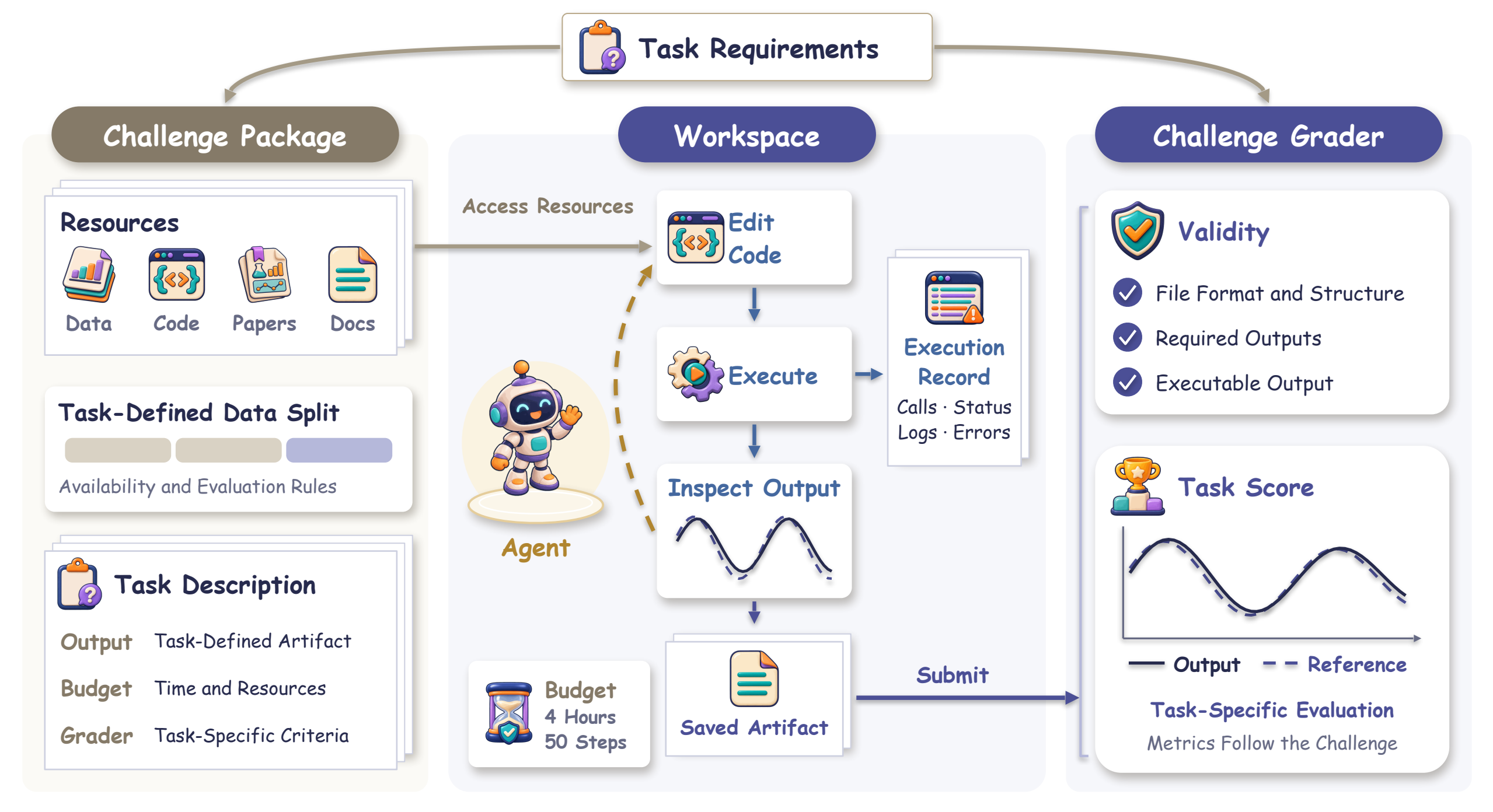
Regression
Error · CRPS



Classification
Accuracy · F1 · Brier



Generation
Rubric · Fidelity · Claims


Task Requirements
Challenge Package
Workspace
Challenge Grader
Resources
Data
Code
Papers
Docs
Task-Defined Data Split
Availability and Evaluation Rules
Task Description
Output
Task-Defined Artifact
Budget
Time and Resources
Grader
Task-Specific Criteria
Access Resources
Edit Code
Execute
Inspect Output
Agent
Execution Record
Calls · Status
Logs · Errors
Budget
4 Hours
50 Steps
Saved Artifact
Submit
Validity
File Format and Structure
Required Outputs
Executable Output
Task Score
Output
Reference
Task-Specific Evaluation
Metrics Follow the Challenge

Question & Series

Question

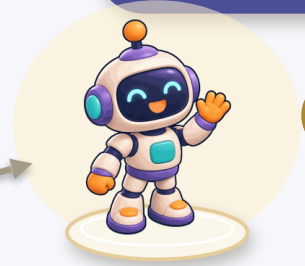
"In which part does the anomaly occur?"

Native Series

Illustrative



Iterative Analysis



Reasoner

Missing Evidence

Next Analytical Action

Detect

Presence & Interval

Observation

Evidence Log

Anomaly Found: Middle

Evidence Gate & Answer

Required Evidence

- ✓ has_anomaly = true
- ✓ anomaly_segment = middle

Both Covered

Answer: Middle

Answer Trace

Native Series → Tool + Arguments → Structured Observation → Answer Evidence

Generalist LLM

Task + Context + Observations



Select Operation & Parameters

Use the Returned Evidence

Operation + Parameters

Numerical Workspace

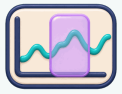
Full-Precision Series

Time	Channel 1	Channel 2
t_1	$x_{1,1}$	$x_{1,2}$
t_2	$x_{2,1}$	$x_{2,2}$

Windows & Metadata Remain Addressable

Temporal Tools

Execute Directly on the Workspace



Slice



Transform



Analyze

Compact Structured Observations

Episodic Memory



Signal Key

Shape & Statistical Structure



Context Key

Task & Textual Context



Retrieve a Successful Episode



Snowstorm → Demand

Rationale + Analytical Trace

Add to the Runtime Tool Set

Experience-Driven Tool Evolution



Executed Traces

Cluster Recurring Procedures



Candidate Tool

Code + Invocation Description



Held-Out Tasks

Meet the Success Threshold



Registered Tool

Available to Later Episodes

Open Question	Controlled Comparison			Evidence to Record
Model or Harness	Native Perception	vs.	Harness Tool	Changed Outputs Verifiability
Memory Under Change	Release v_1 Prior Evidence	→	Release v_2 Revised Values	Revalidation Updated Conclusions
Procedure Transfer	Seen Setting	vs.	Held-Out Setting	Held-Out Performance Reuse Success
Evidence Sensitivity	Evidence Present	vs.	Evidence Removed	Response to Removal Grounded Claims
Prediction to Decision	Forecast (Now)	→	Inventory Action	Forecast Error Action Utility
Projection as Output	Rollout Under A	vs.	Rollout Under B	Agreement with Known Effects
Hindsight Decision	Earlier State	→	Action Replayed	Outcome Change Regret
Probabilistic Outputs	Point Forecast	vs.	Distribution	CRPS Calibration

Match — Backbone · Information Boundary · Resource Budget